BME 423 Lecture Notes - Session 19

Multiple Linear Regression

I. Extending Simple Linear Regression

II. Estimating Trends from a Sample
- A. Multiple Regression Line
- B. Variability about Regression Line
- C. Hypothesis testing of the Model as a Whole
- D. Hypothesis test of Individual Coefficients

III. Hypertension in Families Example – Two Independent Variables

Logistic Regression

I. The Problem

II. Logistic Regression Method

III. Examples
- A. Coronary Heart Disease
- B. O-ring Failure

Old Business

1. What is “Margin of Error” in a survey result report

2. Follow-up of a clinical trial result presented in Glanz and current medical practice

1. What is “Margin of Error” in a survey result report

Size of Difference between Two Means

$$\left(\bar{x}_1 - \bar{x}_2\right) - t_{0.05}(\nu)\cdot s_{\left(\bar{x}_1-\bar{x}_2\right)} < \mu_1 - \mu_2 < \left(\bar{x}_1 - \bar{x}_2\right) + t_{0.05}(\nu)\cdot s_{\left(\bar{x}_1-\bar{x}_2\right)}$$

Size of Difference between two Proportions

$$\left(\hat{p}_1 - \hat{p}_2\right) - z_{0.05}\cdot s_{\left(\hat{p}_1-\hat{p}_2\right)} < p_1 - p_2 < \left(\hat{p}_1 - \hat{p}_2\right) + z_{0.05}\cdot s_{\left(\hat{p}_1-\hat{p}_2\right)}$$

+/- Margin of Error

2. Follow-up of a clinical trial result presented in Glanz and current medical practice

Hormone Replacement Therapy

Grodstein, F. et al. Postmenopausal hormone replacement therapy and mortality. N. Engl J Med. 1997, 336:1769-1775.

5-13 Many postmenopausal women are faced with the decision of whether they want to take hormone replacement therapy or not. Benefits of hormone replacement include decreased risk of cardiovascular disease and osteoporosis. However, hormone replacement therapy has also been associated with increased risk of breast cancer and endometrial cancer. Francine Grodstein and colleagues* investigated the relationship between hormone replacement therapy and overall mortality in a large group of postmenopausal women. The women

TABLE 5-25. Current Use of Hormone Replacement Therapy and Survival

	Number of People	
	Deceased	Alive
Currently using hormone replacement therapy	574	8483
Never used hormone replacement therapy	2051	17,520

H_0 : there is no difference in death rate between the hormone and no-hormone groups

χ^2

$\Rightarrow$ reject H_0 with $P < 0.001$

conclusion: there is a difference in mortality between the hormone and non-hormone groups

$$RR = \frac{n_{TD}/n_T}{n_{ND}/n_N} = \frac{574/9057}{2051/19571} = 0.6047$$

the risk of dying in the hormone group is approximately 60% of the risk of dying in the non-hormone group

2. Follow-up of a clinical trial result presented in Glanz and current medical practice

Hormone Replacement Therapy

osteoporosis. However, hormone replacement therapy has also been associated with increased risk of breast cancer and endometrial cancer. Francine Grodstein and colleagues investigated the relationship between hor-

Follow up on this issue raise by Katie in class

Writing Group for the Women's Health Initiative Investigators. Risks and Benefits of Estrogen Plus Progestin in Healthy Postmenopausal Women: Principal Results From the Women's Health Initiative Randomized Controlled Trial. JAMA. 2002;288(3):321–333. doi:10.1001/jama.288.3.321

Main Outcomes Measures The primary outcome was coronary heart disease (CHD) (nonfatal myocardial infarction and CHD death), with invasive breast cancer as the primary adverse outcome. A global index summarizing the balance of risks and benefits included the 2 primary

Conclusions Overall health risks exceeded benefits from use of combined estrogen plus progestin for an average 5.2-year follow-up among healthy postmenopausal US women. All-cause

2. Follow-up of a clinical trial result presented in Glanz and current medical practice

Hormone Replacement Therapy

"After this research was first published in 2002, the use of hormone therapy fell by 46% within six months – both because clinicians were reluctant to prescribe it and patients were fearful of taking it.

In 2003, the FDA added black box warnings – the most serious warnings, indicating a risk of serious harm or death – to all estrogen-containing hormone products for menopause."

USNews Commentary By The Conversation Feb. 24, 2026, at 2:49 p.m.

https://www.usnews.com/news/health-news/articles/2026-02-24/i-treat-menopause-hormone-replacement-therapy-can-help-heres-the-science-behind-the-fdas-decision-to-remove-warnings

https://www.hhs.gov/press-room/fact-sheet-fda-initiates-removal-of-black-box-warnings-from-menopausal-hormone-replacement-therapy-products.html

2. Follow-up of a clinical trial result presented in Glanz and current medical practice

Hormone Replacement Therapy

FDA U.S. FOOD & DRUG ADMINISTRATION

FDA NEWS RELEASE **For Immediate Release:** February 12, 2026

- Studies have provided evidence that starting HRT within ten years of the onset of menopause can have numerous benefits which for most women outweigh potential risks. Benefits include a reduced risk of all-cause mortality and fractures. HRT has also been associated with 50% reduction in heart attack risk, 64% reduction in cognitive decline, and 35% lower risk of Alzheimer's.

- An analysis of 30 trials with 26,708 women participants found HRT was not associated with increased cancer mortality. In fact, women who start HRT before age 60 appear to have a decreased mortality risk.

https://www.hhs.gov/press-room/fact-sheet-fda-initiates-removal-of-black-box-warnings-from-menopausal-hormone-replacement-therapy-products.html

Cell Phones and Sperm Motility

Fejes et al., Arch Androl, 51:385-393, 2005 – Two groups of cell phone users

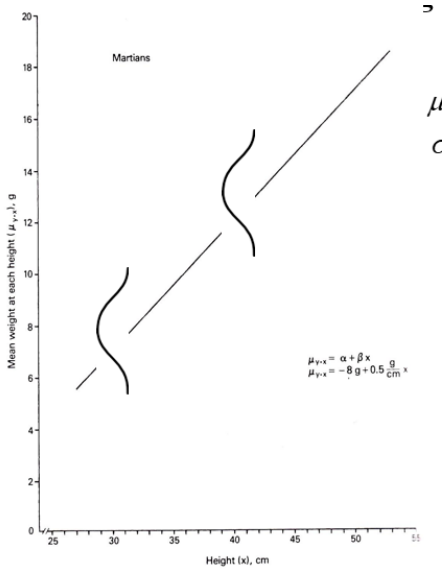
Agarwal, et al., Fertil Steril, 89: 124-128, 2008 – Included a control group of non cell phone users

Conclusion: Sperm cell motility is related to cell phone us.

What is our current understanding?

Simple Linear Regression Summary

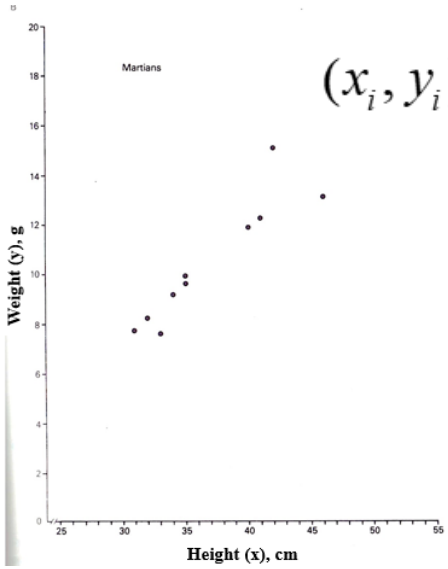
Population



$$\mu_{y|x} = \alpha + \beta x$$

$\sigma_{y|x}$ constant variability about line

Sample



(x_i, y_i) $i = 1, \ldots, n$

$$\hat{y} = a + b \cdot x$$

Estimate of $\mu_{y|x}$

Estimate of α and β

$s^2_{y|x}$ Estimate of $\sigma_{y|x}$

Also

s_a s_b

Simple Linear Regression Summary

Hypothesis Testing:

A. Significance of Trend $H_0 : \beta = 0$

$$t = \frac{b}{s_b} \qquad v = n - 2$$

B. Test for a difference in slopes between two groups $H_0 : \beta_1 = \beta_2$

$$t = \frac{b_1 - b_2}{s_{b_1 - b_2}} \qquad v = n_1 + n_2 - 4$$

Also, for intercepts

Lecture Exercise:

Weight and Height in Males vs. Females Plutonians

Males

Height (inches)	Weight (lbs)
X	Y
62	198
64	207
71	228
73	233
68	217
65	212
64	204
64	201
71	228
67	218
66	210
65	212
68	219
69	226
68	222
73	237
70	222
67	219

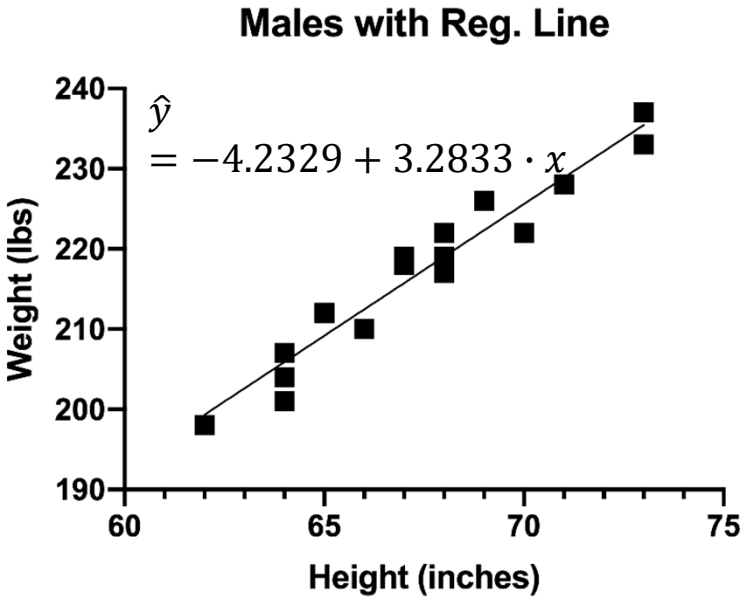
$n_{\mathrm{M}} = 18$

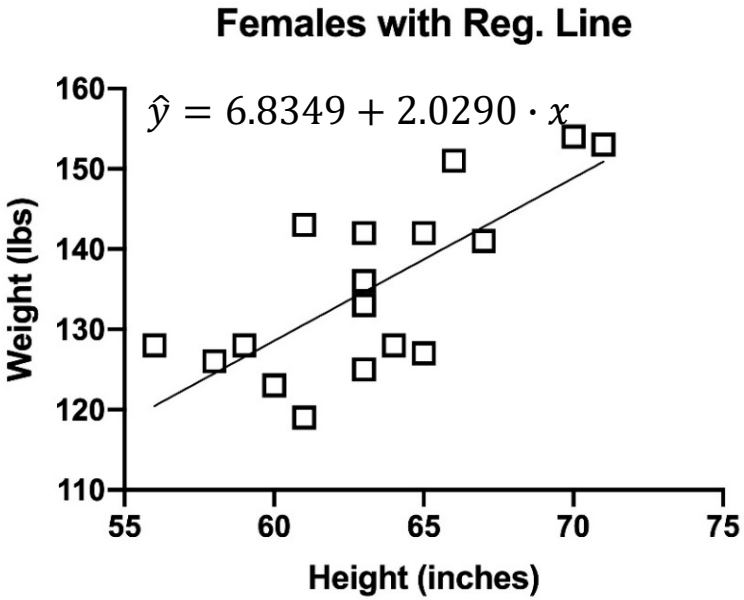
Females

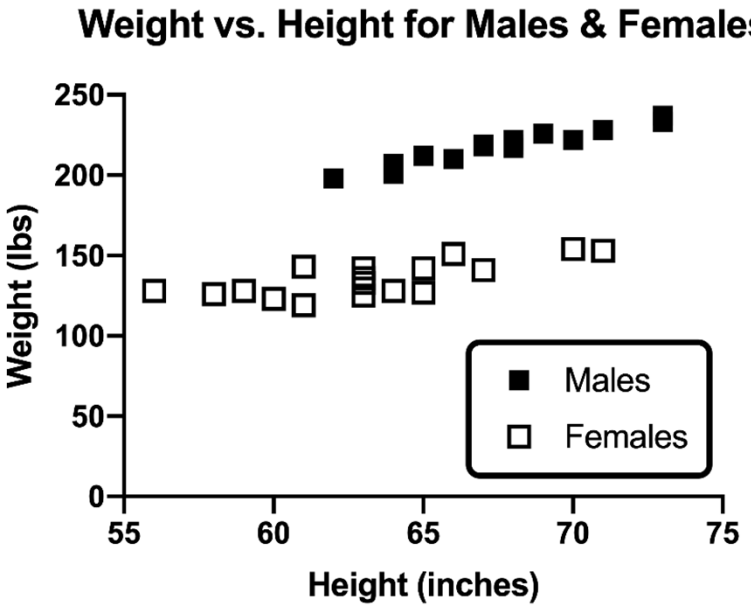
Height (inches)	Weight (lbs)
X	Y
61	119
56	128
64	128
61	143
63	125
71	153
63	142
58	126
63	136
59	128
66	151
67	141
63	133
70	154
65	127
65	142
63	133
60	123

$n_{\mathrm{F}} = 18$

Is the relationship between weight and height for males <u>different</u> than that for females?





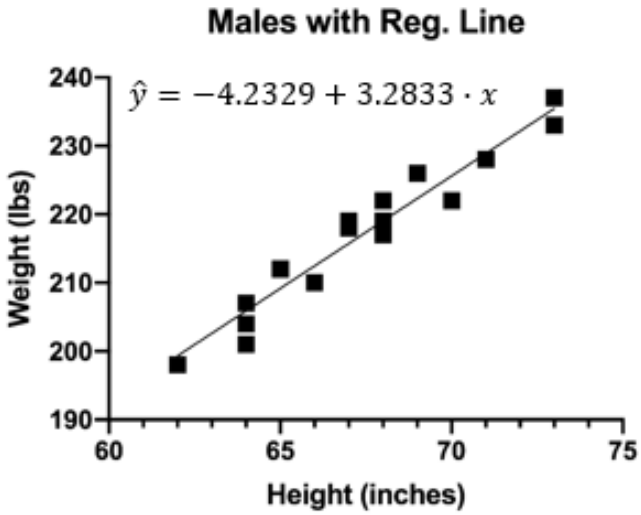


Is the relationship between weight and height for males different than that for females?

Male Plutonians

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -4.2329    13.7901  -0.307    0.763
Height.in     3.2833     0.2041  16.088 2.66e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

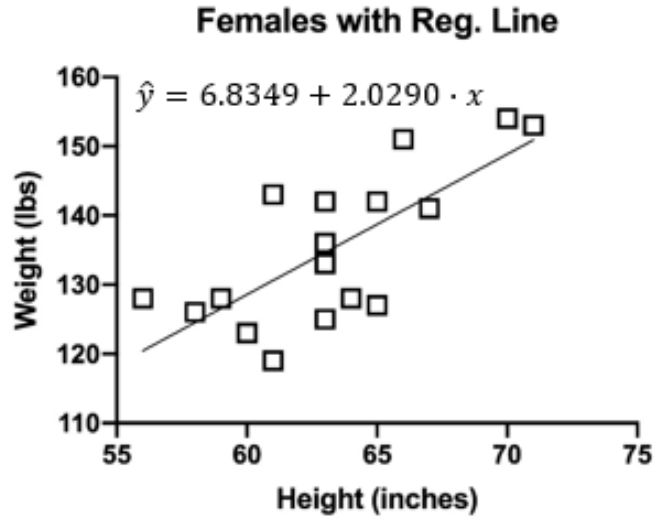
Residual standard error: 2.711 on 16 degrees of freedom
Multiple R-squared:  0.9418,    Adjusted R-squared:  0.9381
F-statistic: 258.8 on 1 and 16 DF,  p-value: 2.661e-11
```



Female Plutonians

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.8349    29.8197   0.229  0.82161
Height.in     2.0290     0.4708   4.309  0.00054 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.491 on 16 degrees of freedom
Multiple R-squared:  0.5372,    Adjusted R-squared:  0.5082
F-statistic: 18.57 on 1 and 16 DF,  p-value: 0.0005401
```



Multiple Linear Regression

I. Extending Simple Linear Regression

More than One Independent Variable – Back to Mars

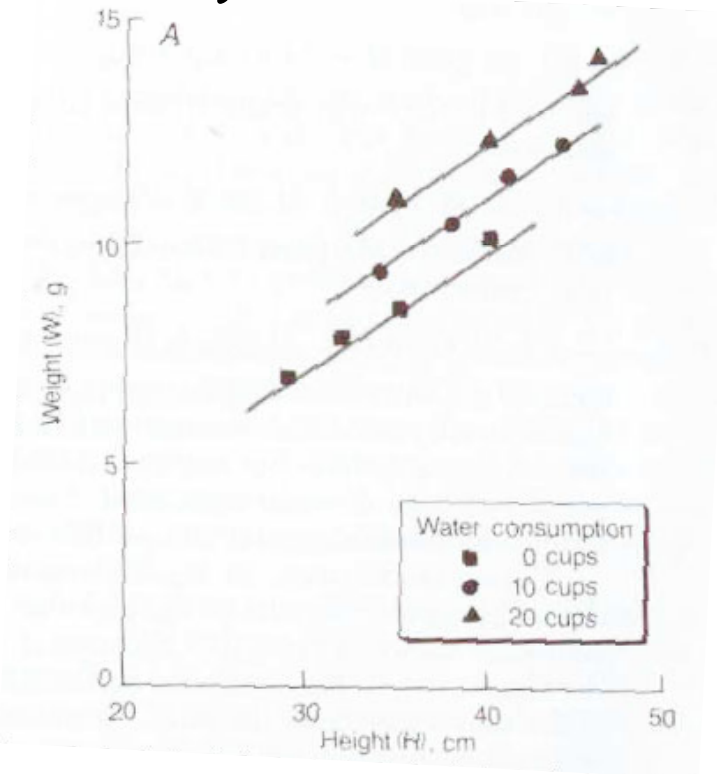
Can we relate Weight to both Height and Water Consumed?

Yes, possible to do via three separate regression analyses:

$$\hat{y} = a + b \cdot x$$

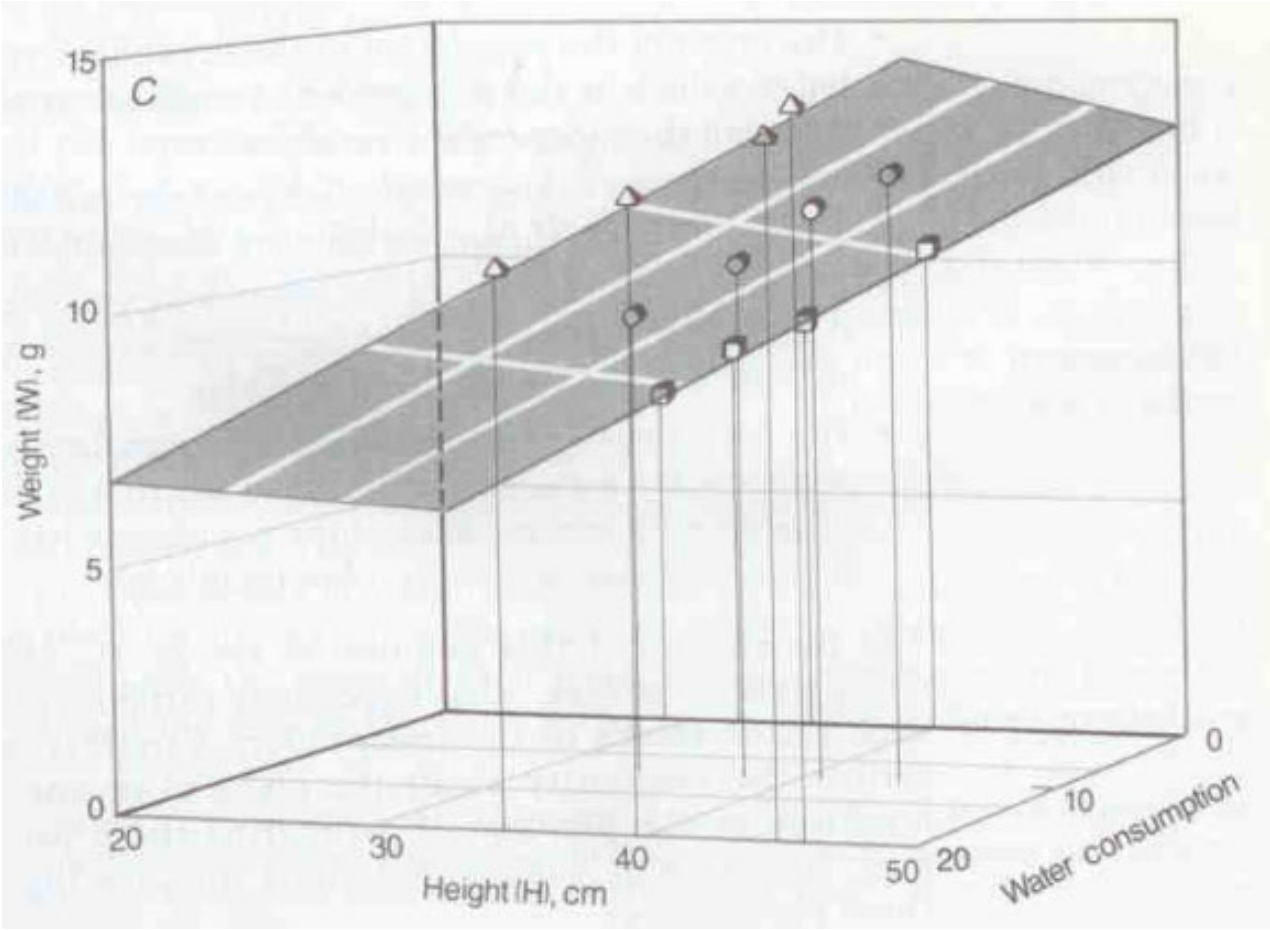
TABLE 1-1 Data of Martian Weight, Height, and Water Consumption

Weight W, g	Height H, cm	Water Consumption C, cups/day
7.0	29	0
8.0	32	0
8.6	35	0
9.4	34	10
10.2	40	0
10.5	38	10
11.0	35	20
11.6	41	10
12.3	44	10
12.4	40	20
13.6	45	20
14.2	46	20



From Glantz, Applied Regression Analysis, 2001.

But, let's look at this more generally:



Best fit <u>plane</u> to the data

II. Estimating Trends from a Sample

A. Multiple Regression Line (two independent variables)

Model of the Data

$$y_i = \beta_0 + \beta_1 \cdot x_{i1} + \beta_2 \cdot x_{i2} + e_i \qquad i = 1,\ldots,n$$

Observation

Independent Variable 1

Independent Variable 2

Error

Regression Line

$$\hat{y} = b_0 + b_1 \cdot x_1 + b_2 \cdot x_2$$

Minimize Sum of Squared Errors over b_0, b_1, b_2

$$SS = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

General Multiple Regression Line

Model of the Data

$$y_i = \beta_0 + \beta_1 \cdot x_{i1} + \beta_2 \cdot x_{i2} + \ldots + \beta_k \cdot x_{ik} + e_i \qquad i = 1, \ldots, n$$

$$y = \begin{bmatrix} y_1 \cdots y_n \end{bmatrix}^T \qquad e = \begin{bmatrix} e_1 \cdots e_n \end{bmatrix}^T \qquad \beta = \begin{bmatrix} \beta_0 \cdots \beta_k \end{bmatrix}^T$$

$$X = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1k} \\ 1 & x_{21} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{bmatrix}$$

$$y = X\beta + e$$

Regression Line

$$\hat{y} = Xb \qquad b = \begin{bmatrix} b_0 \ldots b_k \end{bmatrix}^T$$

Minimize Sum of Squared Errors Over b

$$SS = e^T e = \left(y - \hat{y}\right)^T \left(y - \hat{y}\right)$$

Least Squares Solution

$$\frac{\partial SS}{\partial b} = \frac{\partial}{\partial b}\left(\left(y - Xb\right)^T \left(y - Xb\right)\right) = 0$$

Least Squares Estimate

$$b = \left(X^T X\right)^{-1} X^T y$$

Estimate of model parameters

Sum of Squared Residuals

$$SS_{res} = e^T e = \left(y - Xb\right)^T \left(y - Xb\right)$$

Sum of Squared Errors Evaluated at ***b***

Variability About Regression Line

$$s^2_{y|x} = \frac{SS_{res}}{n - k - 1}$$

Residual Std. Error

$$S_{y|x}$$

Hypothesis testing of the Model as a Whole

Define $SS_{tot} = \left(y - \bar{Y}\right)^T \left(y - \bar{Y}\right)$, where $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} y_i$

$H_0: \beta_1 = \ldots = \beta_k = 0$ versus H_1: at least one $\beta_j \neq 0$

Let: $SS_{mod} = SS_{tot} - SS_{res}$; $MS_{mod} = SS_{mod} / df_{mod}$; $MS_{res} = SS_{res} / df_{res}$

$df_{mod} = k$; $df_{res} = n - k - 1$

$$F = \frac{MS_{mod}}{MS_{res}}; \quad \text{where } F \sim F_{\alpha}\left(df_{\text{mod}}, df_{res}\right)$$



Hypothesis testing of Individual Coefficients using Standard Errors of the Coefficients

$$H_0: \beta_j = 0 \quad \text{versus} \quad H_1: \beta_j \neq 0$$

$$t = \frac{b_j}{s_{b_j}}; \quad \text{where } t \sim t_\alpha\left(n-k-1\right)$$



if $|t| > t_{crit} = t_\alpha(\nu)$ *reject* H_0 *with* $P < \alpha$
(using two-tailed t table)

$$\text{where } s_{b_j}^2 = s_{y|x}^2 diag\left\{\left(X^T X\right)^{-1}\right\}$$

Correlation Coefficient ($R = r$)

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$



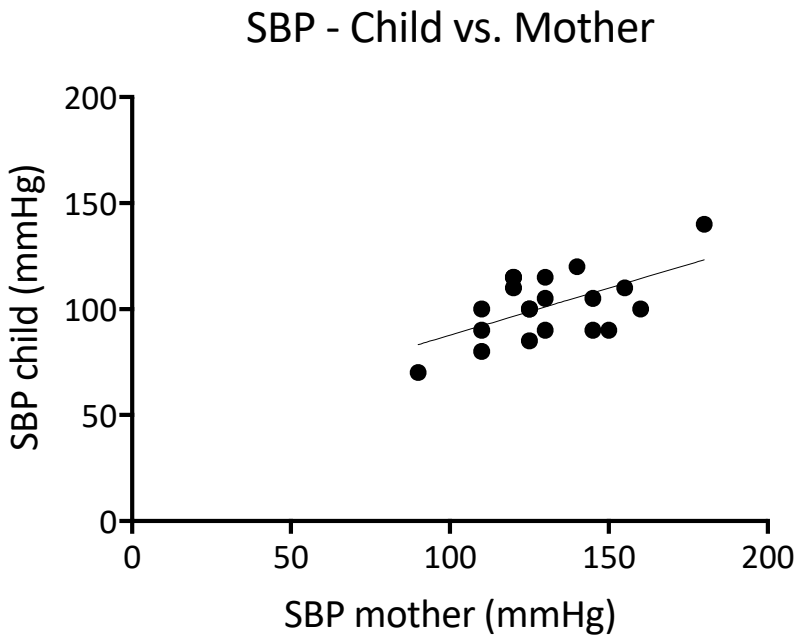
III. Hypertension in Families Example – Two Independent Variables

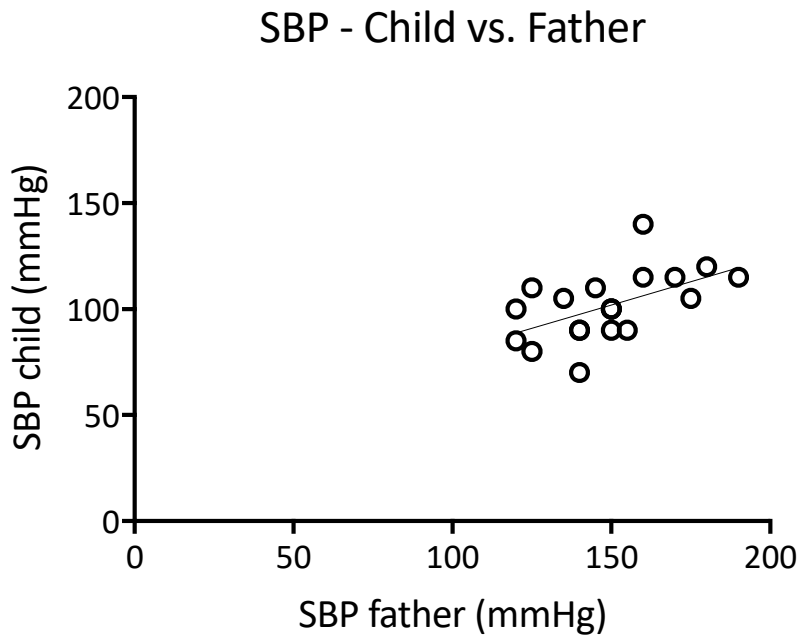
Much research has been devoted to the etiology of hypertension. One general problem is to determine to what extent hypertension is a genetic phenomenon. This issue ws examined in 20 families by measuring the SBP of the mother, father, and first-born child in the family.

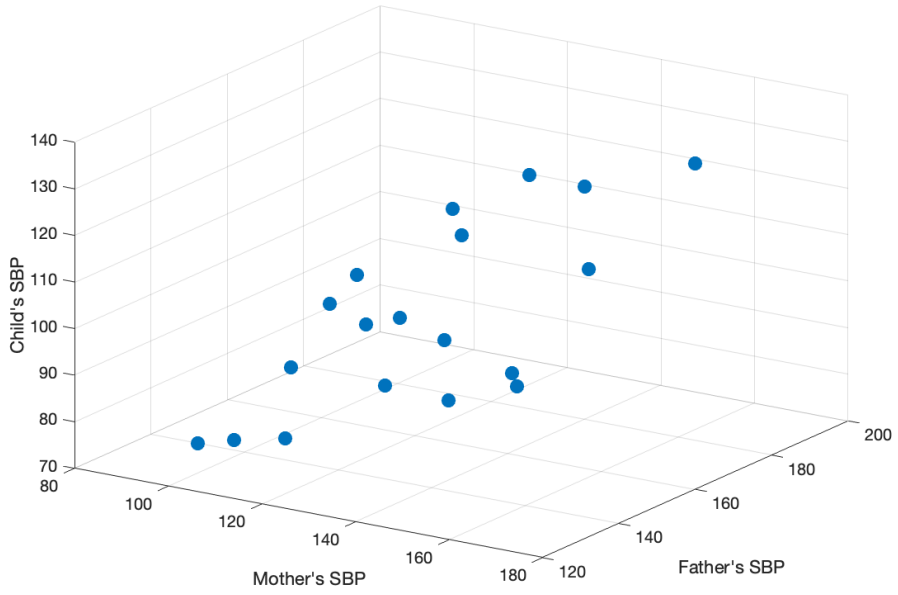
From Rosner, Fundamentals of Biostatistics, 2006.

Family	SBP child (mm Hg), t	SBP mother (mm Hg), y	SBP father (mm Hg), x
1	90	130	140
2	85	125	120
3	120	140	180
4	100	110	150
5	105	145	175
6	100	160	120
7	110	120	145
	■ ■ ■		
18	110	155	125
19	90	110	140
20	100	125	150

Hypertension in Families: Exploratory Analysis of the Data







III. Hypertension in Families Example – Two Independent Variables

Perform a multiple linear regression between SBP of the child (y) and SBP of the mother (x_1) together with the SBP of the father (x_2) .

SBPchildparent.R

Source on Save

```
1  BPchildparent <- read_csv("D:/BME423/BPchildparent.csv")
2
3  fit <- lm(SBPchild ~ SBPmother+SBPfather, data = BPchildparent)
4  summary(fit)
5
```

Family	SBPchild	SBPmother	SBPfather
1	90	130	140
2	85	125	120
3	120	140	180
4	100	110	150
5	105	145	175
6	100	160	120
7	110	120	145
8	140	180	160
9	115	120	190
10	105	130	135
11	100	125	150
12	80	110	125
13	70	90	140
14	115	120	170
15	90	150	150
16	90	145	155
17	115	130	160
18	110	155	125
19	90	110	140
20	100	125	150

```
> fit <- lm(SBPchild ~ SBPmother+SBPfather, data = BPchildparent)
> summary(fit)

Call:
lm(formula = SBPchild ~ SBPmother + SBPfather, data = BPchildparent)

Residuals:
     Min       1Q   Median       3Q      Max
-19.9434  -3.8354   0.8644   8.5785  14.8080

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -15.6892    23.6503  -0.663  0.51598
SBPmother     0.4225     0.1185   3.565  0.00238 **
SBPfather     0.4150     0.1248   3.325  0.00401 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.67 on 17 degrees of freedom
Multiple R-squared:  0.5973,    Adjusted R-squared:  0.5499
F-statistic: 12.61 on 2 and 17 DF,  p-value: 0.0004388
```

$$\hat{y} = b_0 + b_1 x_1 + b_2 x_2 = -15.7 + 0.423 x_1 + 0.415 x_2$$

$$s_{y|x} = 10.67$$

The regression line between SBP of child and both mother and father

$$\hat{y} = -15.7 + 0.423x_1 + 0.415x_2, \quad s_{y|x} = 10.67$$

Based on this multiple regression result, test the following hypothesis:

$$H_0: \beta_1 = \beta_2 = 0 \text{ versus } H_1: \beta_1 \neq 0 \text{ and/or } \beta_2 \neq 0$$

$$F = \frac{MS_{mod}}{MS_{res}}; \quad \text{where } F \sim F_{\alpha}\left(df_{\text{mod}}, df_{res}\right) \qquad df_{mod} = k; \quad df_{res} = n - k - 1$$

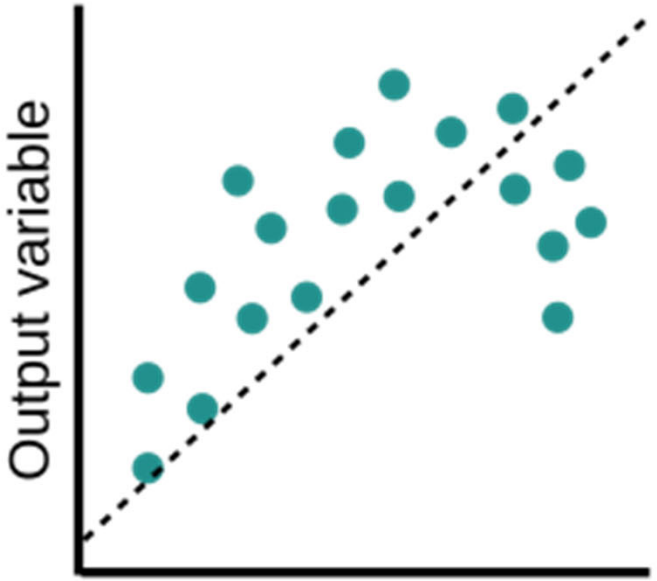
```
Residual standard error: 10.67 on 17 degrees of freedom
Multiple R-squared:  0.5973,    Adjusted R-squared:  0.5499
F-statistic: 12.61 on 2 and 17 DF,  p-value: 0.0004388
```

Thus, we conclude with $P < 0.001$ that the mother's and father's SBPs, when considered together, are significant predictors of the child's SBP (i.e., we reject the null hypothesis)

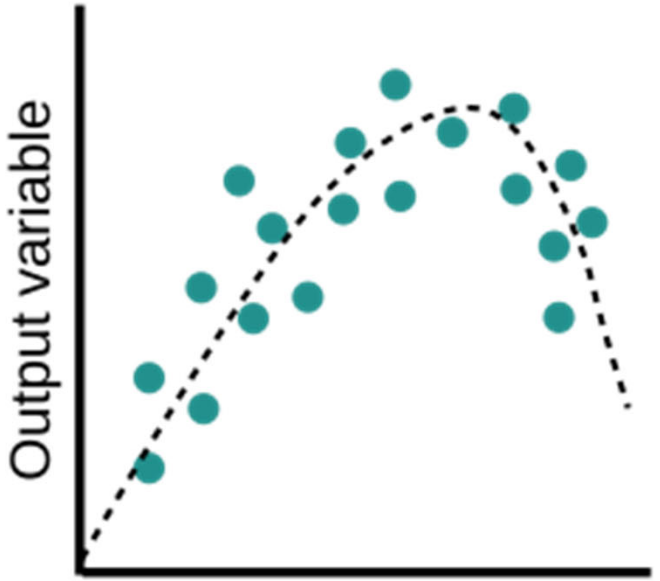
A Note About Model Complexity

- Increasing number of parameters in a model will generally improve the model fit to the observed data
- However, at some point, the added complexity ceases to provide useful, generalizable information about the system (i.e., you start fitting your model to the noise in the data instead of illuminating the nature of the system you are trying to model)
- Need to balance making model complex enough to handle the nature of the data without starting to "track" the noise in the data

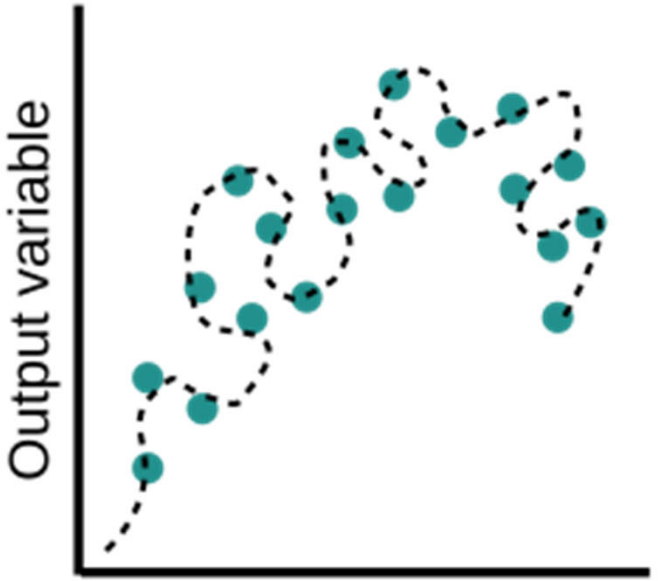
Underfitted



Good Fit/Robust



Overfitted



Logistic Regression

I. The Problem

How is Coronary Heart Disease Related to Age?

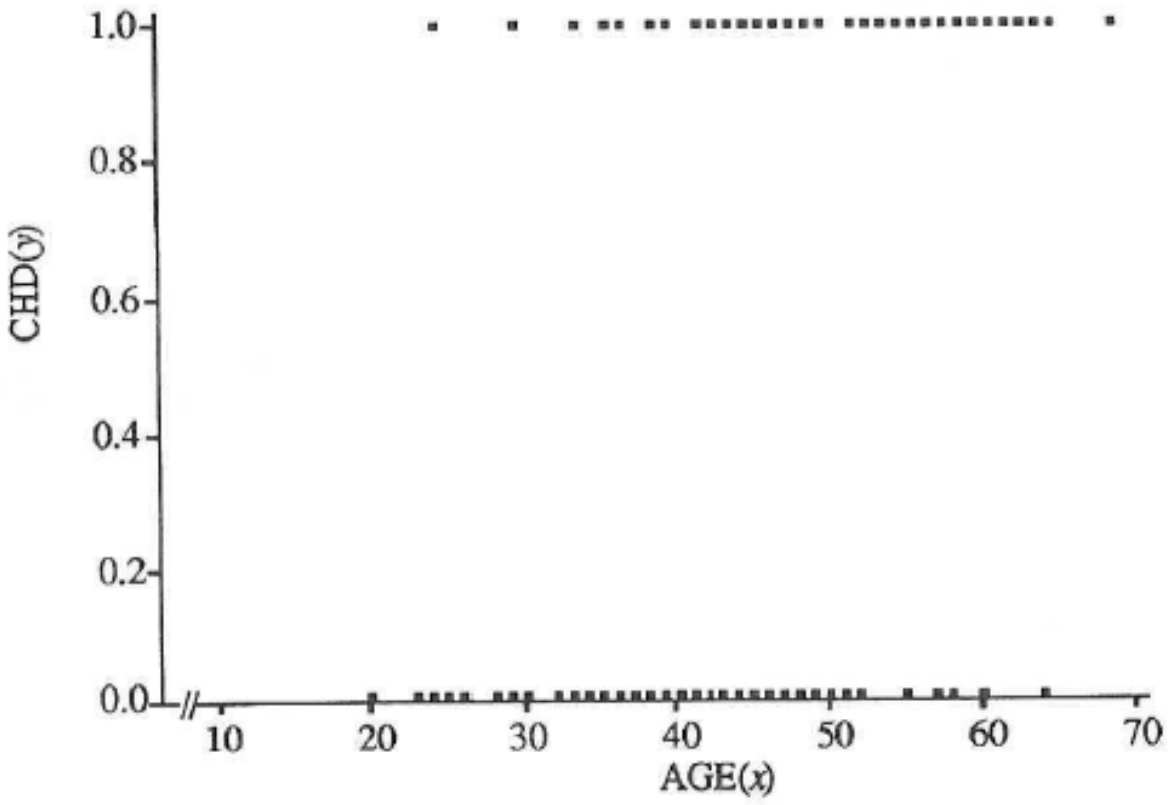


Figure 1.1 Plot of CHD by Age.

Presence of CHD is a categorical dependent variable: the patient either has CHD or does not

A linear regression relationship between the presence of CHD and Age is not appropriate

Can we use these data to find a relationship between the probability of getting CHD and age?

Hosmer & Lemeshow, Applied Logistic Regression. 1989

How is Coronary Heart Disease Related to Age?

Table 1.1 Age and Coronary Heart Disease Status (CHD) of 100 Subjects.

ID	AGRP	AGE	CHD	ID	AGRP	AGE	CHD	ID	AGRP	AGE	CHD
1	1	20	0	35	3	38	0	68	6	51	0
2	1	23	0	36	3	39	0	69	6	52	0
3	1	24	0	37	3	39	1	70	6	52	1
4	1	25	0	38	4	40	0	71	6	53	1
31	3	37	0	65	5	49	1	98	8	64	1
32	3	37	1	66	6	50	0	99	8	65	1
33	3	37	0	67	6	50	1	100	8	69	1
34	3	38	0								

Consider constructing a frequency table based on age groups

Table 1.2 Frequency Table of Age Group by CHD.

		CHD		
Age Group	n	Absent	Present	Mean (Proportion)
20 – 29	10	9	1	0.10
30 – 34	15	13	2	0.13
35 – 39	12	9	3	0.25
40 – 44	15	10	5	0.33
45 – 49	13	7	6	0.46
50 – 54	8	3	5	0.63
55 – 59	17	4	13	0.76
60 – 69	10	2	8	0.80
Total	100	57	43	0.43

Hosmer & Lemeshow, Applied Logistic Regression. 1989

Table 1.2 Frequency Table of Age Group by CHD.

Age	Age Group	n	CHD Absent	CHD Present	Mean (Proportion)
24.5	20 – 29	10	9	1	0.10
32	30 – 34	15	13	2	0.13
37	35 – 39	12	9	3	0.25
42	40 – 44	15	10	5	0.33
47	45 – 49	13	7	6	0.46
52	50 – 54	8	3	5	0.63
57	55 – 59	17	4	13	0.76
64.5	60 – 69	10	2	8	0.80
	Total	100	57	43	0.43

Sample mean (proportion) given x – Prob of CHD

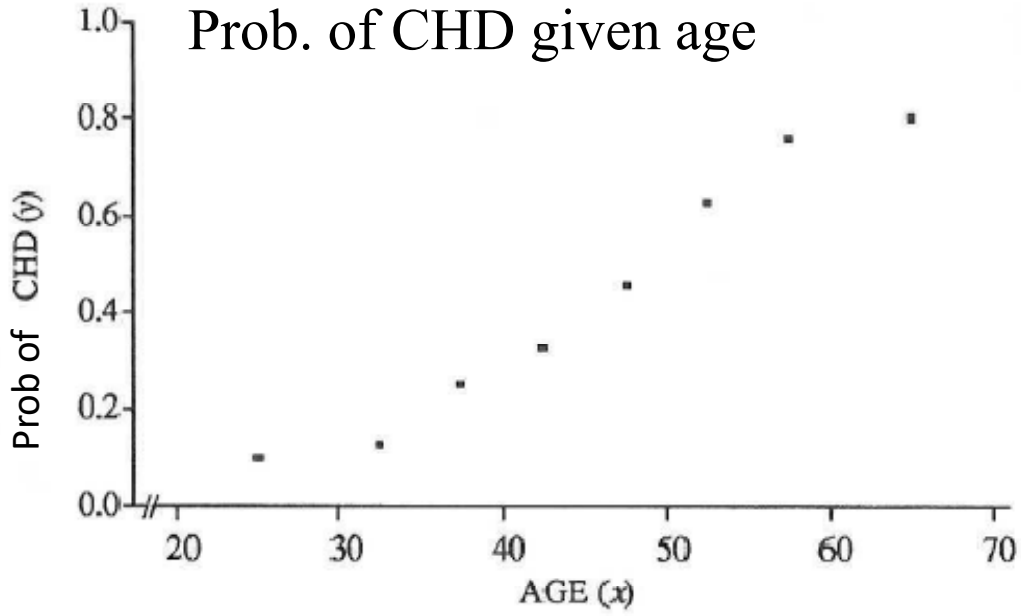


Figure 1.2 Plot of the Mean of CHD in Each Age Group.

Consider two approaches:

1. Linear regression of Prob CHD vs Age

2. Use the Logistic function to relate Prob of CHD to Age

Logistic function: $x \in \left[-\infty,+\infty\right]$ with $y \in \left[0,1\right]$

$$p_i = \frac{e^{(\beta_0+\beta_1 \cdot x_i)}}{1+e^{(\beta_0+\beta_1 \cdot x_i)}} = \frac{1}{1+e^{-(\beta_0+\beta_1 \cdot x_i)}}$$

p_i is probability associated with x_i

It is a flexible function whose shape is defined by β_0 and β_1

How to use the Logistic function to relate Prob of CHD to Age

1. Transform the Logistic function as follows:

$$p_i = \frac{e^{(\beta_0+\beta_1\cdot x_i)}}{1+e^{(\beta_0+\beta_1\cdot x_i)}} \quad \Rightarrow \quad \frac{p}{1-p} = e^{(\beta_0+\beta_1\cdot x)}$$

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \cdot x \quad \Leftarrow \quad \underline{\text{Logit Transformation:}}\ \text{logit}(p)$$

2. Perform a Logit transformation of the data:

AGE	p	logit(p)
24.5	0.1	−0.9542
32	0.13	−0.8256
…	…	…
64.5	0.8	0.6021

3. Do a linear regression using logit(p) data to estimate β_0 and β_1

Using the lm() function in R to regress logit(p) data on Age

```
> ##4. Linear Regression
> regression <- lm(LogitP ~ AGE, data=DATA)
> summary(regression)

Call:
lm(formula = LogitP ~ AGE, data = DATA)

Residuals:
      Min        1Q    Median        3Q       Max
-0.123404 -0.058443 -0.006651  0.080965  0.114054

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.095787   0.129467  -16.19 3.53e-06 ***
AGE          0.043550   0.002802   15.54 4.49e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09857 on 6 degrees of freedom
Multiple R-squared:  0.9758,    Adjusted R-squared:  0.9717
F-statistic: 241.6 on 1 and 6 DF,  p-value: 4.49e-06
```

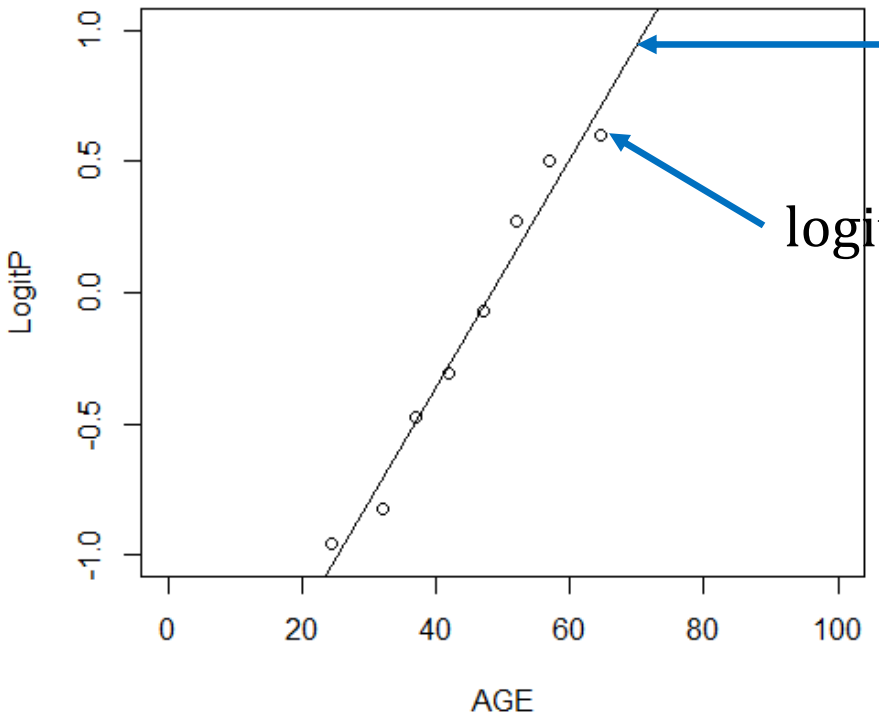
b_0

b_1

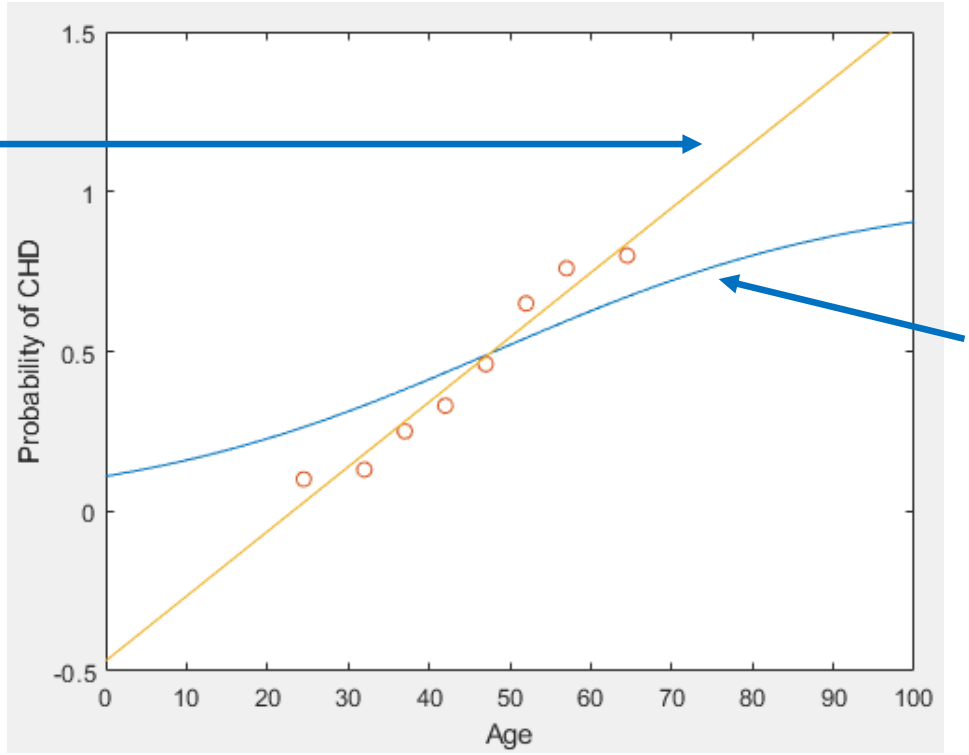
Resulting regression plot using logit(p) data

$$\text{logit}(p) \equiv \ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 \cdot Age$$

$b_0 = -2.0958$ $\quad$ $b_1 = 0.04355$



Linear regression of p on Age



Plot of Prob of CHD (p) using logistic function with

$b_0 = -2.0958$

$b_1 = 0.04355$

II. The Method of Logistic Regression

- The logistic regression framework can be extended to the case of multiple independent (explanatory) variables, which has important application to the classification problem in machine learning

$$p_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot x_{1,i} + \cdots + \beta_k \cdot x_{k,i})}}$$

$$\text{logit}(p) = \beta_0 + \beta_1 \cdot x_1 + \cdots + \beta_k \cdot x_k$$

- Potential issues with $\text{logit}(p)$ method: we may not have enough data to calculate probabilities as we did on previous slides; choice of intervals for calculating probabilities arbitrarily influences results; logit transformation of data can distort results

- We can estimate the parameters of the logistic function β_0, β_1, etc. without explicitly calculating probabilities by using the method of maximum likelihood estimation (we will not present the details behind this method)

III. Example

A. How is Coronary Heart Disease Related to Age?

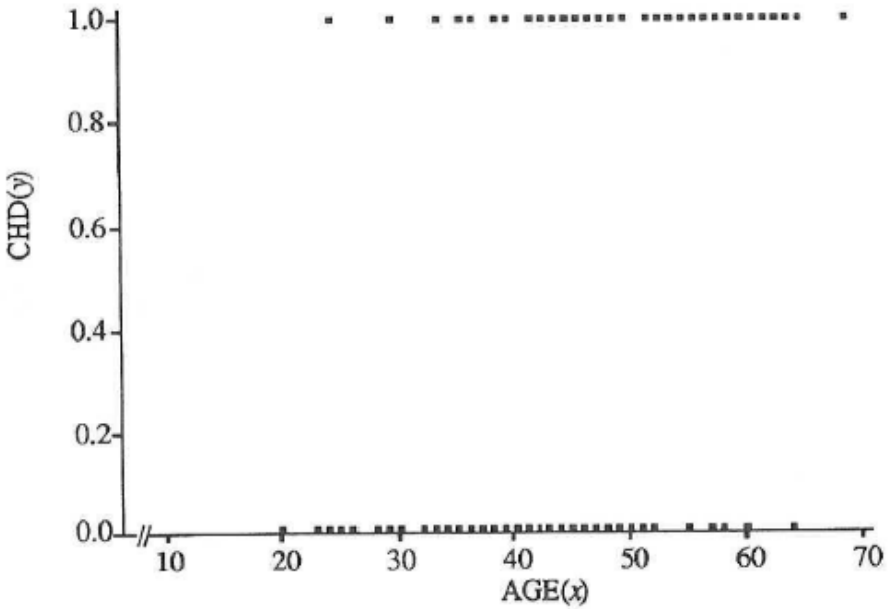


Figure 1.1 Plot of CHD by Age.

Table 1.1 Age and Coronary Heart Disease Status (CHD) of 100 Subjects.

ID	AGRP	AGE	CHD	ID	AGRP	AGE	CHD	ID	AGRP	AGE	CHD
1	1	20	0	35	3	38	0	68	6	51	0
2	1	23	0	36	3	39	0	69	6	52	0
3	1	24	0	37	3	39	1	70	6	52	1
4	1	25	0	38	4	40	0	71	6	53	1
31	3	37	0	65	5	49	1	98	8	64	1
32	3	37	1	66	6	50	0	99	8	65	1
33	3	37	0	67	6	50	1	100	8	69	1
34	3	38	0								

Use logistic regression to estimate

$$\beta_0 \ \& \ \beta_1$$

$$p_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot x_i)}}$$

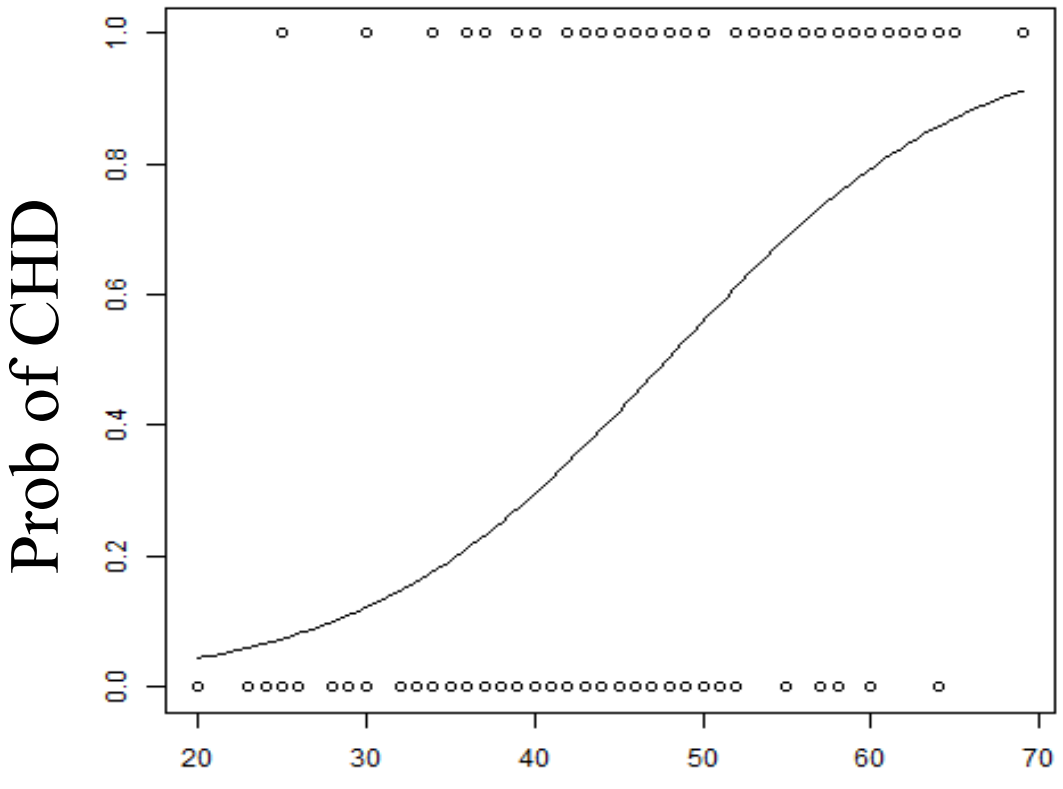
```
> ## 2. Logistic Regression
> logistic <- glm(CHD ~ AGE, data=DATA, family="binomial")
> summary(logistic)

Call:
glm(formula = CHD ~ AGE, family = "binomial", data = DATA)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -5.30945    1.13365  -4.683 2.82e-06 ***
AGE          0.11092    0.02406   4.610 4.02e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

b_0

b_1



$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 \cdot x)}}$$

Interpretation of logistic regression based on the idea of the *odds* that an event might occur

$$odds = \frac{probability\ event\ occurring}{probability\ event\ not\ occurring} = \frac{p}{1-p}$$

$$p = \frac{1}{1+e^{-(\beta_0+\beta_1 \cdot x)}} \qquad odds = \frac{1/\left(1+e^{-(\beta_0+\beta_1 \cdot x)}\right)}{1-1/\left(1+e^{-(\beta_0+\beta_1 \cdot x)}\right)} = e^{(\beta_0+\beta_1 \cdot x)} = e^{\beta_0} e^{\beta_1 \cdot x}$$

For someone age $= 60$

probability of having CHD $\quad p_i = \frac{1}{1+e^{-(-5.30945+0.11092 \cdot 60)}} = 0.79$

odds of having CHD $\quad odds = \frac{p}{1-p} = \frac{0.79}{1-0.79} = 3.76$

we can say the odds of having CHD for a 60-year-old are 3.76: 1

III. Example

B. How is O-Ring Failure Related to Temperature?

- Data were collected on the incidence of failures for O-rings in 24 shuttle launches leading up to the *Challenger* disaster in January 1986
- Launch temperatures were also collected for each event

Temperature	Failure	Temperature	Failure	Temperature	Failure
53	Y	68	N	75	N
56	Y	69	N	75	Y
57	Y	70	N	76	N
63	N	70	Y	76	N
66	N	70	Y	78	N
67	N	70	Y	79	N
67	N	72	N	80	N
67	N	73	N	81	N

(J. L. Devore, Prob. and Stats. for Engineers and Scientists, 2008 *)*

Temp	Failure
53	1
56	1
57	1
63	0
66	0
67	0
78	0
79	0
80	0
81	0

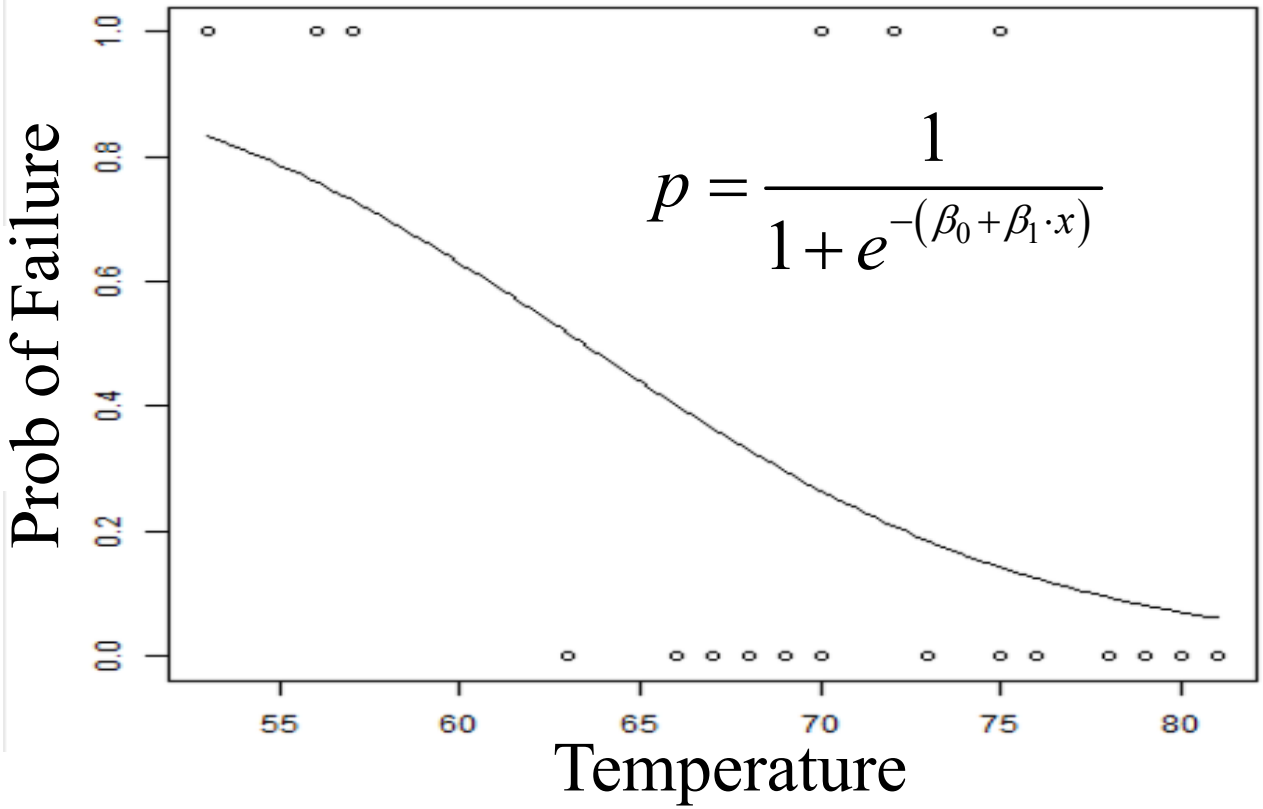
```
> ## 2. Logistic Regression
> logistic <- glm(Failure ~ Temp, data=DATA, family="binomial")
> summary(logistic)

Call:
glm(formula = Failure ~ Temp, family = "binomial", data = DATA)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  9.83198    5.39833   1.821   0.0686 .
Temp        -0.15500    0.07883  -1.966   0.0493 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

b_0

b_1



Summary of Some Statistical Methods to Test Hypotheses

Scale of measurement	Type of experiment: Two treatment groups consisting of different individuals	Three or more treatment groups consisting of different individuals	Before and after a single treatment in the same individuals	Multiple treatments in the same individuals	Association between two variables
Interval (and drawn from normally distributed populations*)	Unpaired *t* test (Chapter 4)	Analysis of variance (Chapter 3)	Paired *t* test (Chapter 9)	Repeated-measures analysis of variance (Chapter 9)	Linear regression, Pearson product-moment correlation, or Bland-Altman analysis (Chapter 8)
Nominal	z Test Chi-square analysis-of-contingency table (Chapter 5)	Chi-square analysis-of-contingency table (Chapter 5)	McNemar's test (Chapter 9)	Cochrane Q†	Relative risk or odds ratio (Chapter 5) Logistic regression (slides)
Ordinal‡	Mann-Whitney rank-sum test (Chapter 10)	Kruskal-Wallis statistic (Chapter 10)	Wilcoxon signed-rank test (Chapter 10)	Friedman statistic (Chapter 10)	Spearman rank correlation (Chapter 8)
Survival time	Log-rank test or Gehan's test (Chapter 11)				

*If the assumption of normally distributed populations is not met, rank the observations and use the methods for data measured on an ordinal scale.

†Not covered in this text.

‡Or interval data that are not necessarily normally distributed.